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X-EUROPE 5

Scenery for X-Plane 10 and 11

www.simheaven.com

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X-EUROPE is a scenery using XP11 objects like the standard scenery, but the objects are not set by an auto-gen rule, they are set following OSM data, which leads to a more realistic look and feel.

You could see results of different demo flights with older versions of **X-EUROPE** on YouTube, even a comparism between various scenery configurations (Ortho4XP, Standard, **X-EUROPE**):

<https://www.youtube.com/channel/UCZzxqygX3ZRF5rIlZuEE7RA/videos>

This is the fifth step bringing localized objects for countries and regions, now including Northern Germany, UK and Ireland, Mediterranean, the Alps region, Norway/Scandinavia and Greece.

Delete older versions, copy all 7 folders from the archive to the Custom Scenery folder, after starting and ending X-Plane, you need to adjust the order of entries in scenery_packs.ini, refer page 6 for details.

The complete scenery consists of 7 layers (see page 7 for description):

- simHeaven_X-Europe-1-vfr
- simHeaven_X-Europe-2-regions
- simHeaven_X-Europe-3-extras
- simHeaven_X-Europe-4-scenery
- simHeaven_X-Europe-5-forests
- simHeaven_X-Europe-6-autogen
- simHeaven_X-Europe-7-network

“simHeaven_Vegetation_Library” has to be installed!

<https://simheaven.com/simdownloads/vegetation-library/>

OpenSceneryX library must be installed and updated!

<https://www.opensceneryx.com/>

For ortho sceneries you may change library.txt! (see page 8)

If you like the scenery and want to support my work, think about a donation to simHeaven:

Donate to simHeaven: https://www.paypal.com/donate?hosted_button_id=RJRM2YX6R928Q
or use paypal.me/simheaven and „Send2Friend“ to avoid PayPal fees

Have fun flying!

Armin (aka PilotBalu)

simheaven.com

Changelog

v5.3

- 1-vfr 24 new objects, 2 corrected
- 2-regions farm facades adapted to regions
- 4-scenery completely new (OSM data Nov 17th 2020)
height now accurate for objects and buildings
complex buildings (building:part in OSM) added
tanks and wastewater bassins are complete now
default height for wind turbines (no height in OSM) set to 50m
some towers (missing tower:type in OSM) not generated
public.fac improved, Schiff.obj scaled

v5.2.1

- 1-vfr Germany Burg Hohenzollern corrected
France Grenoble Stade de Alpes added
Belgium Leper St_Martin_s_Cathedral white space corrected

v5.2

- 1-vfr 278 new objects added (mainly GB and Eastern Europe, see list)
- 2-regions folder objects cleaned up
- 3-extras completely new (OSM data September 25th 2020)
vegetation at underground water eliminated
- 4-scenery completely new (OSM data October 17th 2020)
Fly Agis asynchronous wind-turbines added
generation of building:part objects slightly changed
tiles +50+002, +51+004, +51+005 now "OSM only"
folder objects cleaned up
- 5-forests updated tile +47+007 (was missing forest near Grenchen)
- 6-autogen tiles +50+002, +51+004, +51+005 removed
- 7-network completely new (OSM data September 25th 2020)
generation of power lines slightly changed

v5.1

- 2-regions parking facade corrected (has lead to errors when renaming folders)
- 4-scenery 170 updated tiles +43+006 to +54+022 (only OSM area, OSM data June 21th 2020)

V5.0

- initial release

For a detailed changelog since v3.0, please see changelog.txt in 4-scenery

Credits

- Austin Meyer, Laminar Research, for the permission to use [X-Plane11](#) objects and textures
- Tony Wroblewski for creating [world2plane](#)
- [OpenStreetMap](#) and all editors
- [GeoFabrik](#) for providing the downloads
- [R2 Library](#) authors for some used objects
- Austin Goudge for implementing simHeaven entries into [OpenSceneryX](#) library
- Agnes (FlyAgi) for integration of global vegetation
- Andras (alpilotx) for permission to use forests from [HD Scenery Mesh v4](#)
- Didier (Oscar Pilote, [Ortho4XP](#)) for the script to split forests
- Scottish Wings for objects from [The Handy Objects Library](#)
- Georgios (thasos/discordia) and Chris (chrisag) for implementing Greece
- Nicola (nicotum-xp) for implementing [Four Seasons](#)
- Dieter (diego) for Wachau Austria objects
- Sven (Sven S.), Peter (Pitt153), Axel (Wildkatz), and Roland (maydayc) for their massive help
- all donators and supporters, thanks to all 😊

Description

Due to #stayathome times, we spent some time doing the next full version, mainly changed layer scenery and forests, added region Germany North and UK/Ireland, changed a lot in generation of each layer, removed service streets (at airports) and done general improvements.

Thanks to all! 😊

This is the fifth step bringing localized objects for countries and regions, now including new the Northern Germany, UK and Ireland, the Mediterranean area, the Alps region, Norway/Scandinavia and Greece, so you will recognize where you fly if you simply look outside the window. Hopefully other country packages for e.g. France, Denmark etc. may follow if I get support from community. The country packages will change objects in this country or region to typical objects.

This scenery was generated with world2xplane beta 0.7.4, a tool made by Tony Wroblewski.

OSM data is provided by <http://geofabrik.de> - many users all over the world are doing the input for free. Please help and improve your region, visit openstreetmap.org for details or support the project at <https://donate.openstreetmap.org/>

OSM data coverage and quality

This is the coverage of X-Europe, the same coverage as the download file from Geofabrik plus Canary Islands.

It ends in the East at lon 45.



Even if OSM data is quite good in Europe, it isn't good enough in some regions.

In **X-EUROPE** there are regions based only on OSM data and others based on OSM data plus added autogen objects or forests. Smart exclusions allow or disable autogen objects in these regions where OSM data is poor.

At the right is an overview where only OSM data is used (yellow), outside this area you will see sometimes additional objects (magenta)



Sceneries and libraries needed

For the **X-EUROPE** you only need **one** scenery for the terrain (which includes mesh and terrain textures), this could be either

- the standard X-Plane scenery, i.e. the scenery of X-Plane 11,
- HD/UHD Mesh scenery or any other improved terrain scenery or
- orthographic sceneries, made using G2XPL or Ortho4XP, which is the preferred option as the ground perfectly fits to the objects. If so, please change library.txt (page 8)

One of them is only needed, not all of them! 😊

Additionally, you have to **install or update the OpenSceneryX** library, see <http://opensceneryx.com> as some objects are referenced by this library.

And you **now need the overlays for Ortho** sceneries (if installed) for forests in OSM+autogen areas!

scenery_packs.ini

For a better overview of your scenery_packs.ini, you may add separators, i.e. empty folders named like "----- my sceneries -----" as you see here:

```
SCENERY_PACK C:\-----\----- Custom Airports -----/
SCENERY_PACK Custom Scenery/_EDDN Nuernberg/
SCENERY_PACK Custom Scenery/_EDQH Herzogenaurach/
SCENERY_PACK C:\-----\----- Aerosoft Sceneries -----/
SCENERY_PACK Custom Scenery/Aerosoft - EDDF Frankfurt/
SCENERY_PACK Custom Scenery/Global Airports/
SCENERY_PACK Custom Scenery/KSEA Demo Area/
SCENERY_PACK Custom Scenery/LOWI Demo Area/
SCENERY_PACK Custom Scenery/X-Plane Landmarks - London/
SCENERY_PACK C:\-----\----- VFR Objects -----/
SCENERY_PACK F:\X-Plane_Scenery\VFR-Objects\VFR-Objects_+49+010_DE_Ansbach/
SCENERY_PACK F:\X-Plane_Scenery\VFR-Objects\VFR-Objects_+49+011_DE_Erlangen/
SCENERY_PACK C:\-----\----- X-EUROPE -----/
SCENERY_PACK Custom Scenery/simHeaven_Vegetation_Library/
SCENERY_PACK Custom Scenery/simHeaven_X-Europe-1-vfr/
SCENERY_PACK Custom Scenery/simHeaven_X-Europe-2-regions/
SCENERY_PACK Custom Scenery/simHeaven_X-Europe-3-extras/
SCENERY_PACK Custom Scenery/simHeaven_X-Europe-4-scenery/
SCENERY_PACK Custom Scenery/simHeaven_X-Europe-5-forests/
SCENERY_PACK Custom Scenery/simHeaven_X-Europe-6-autogen/
SCENERY_PACK Custom Scenery/simHeaven_X-Europe-7-network/
SCENERY_PACK C:\-----\----- Libraries -----/
SCENERY_PACK Custom Scenery/OpenSceneryX/
SCENERY_PACK C:\-----\----- Ortho4XP Sceneries -----/
SCENERY_PACK F:\X-Plane_Scenery\o4xp\yOrtho4XP_Overlays/
SCENERY_PACK F:\X-Plane_Scenery\o4xp\z+41+002_ES_Barcelona_obx17+/
SCENERY_PACK F:\X-Plane_Scenery\o4xp\z+49+010_DE_Ansbach_obx17+/
SCENERY_PACK F:\X-Plane_Scenery\o4xp\z+49+011_DE_Nuernberg_obx17+/
```

To ensure the best results, entries in the scenery_packs.ini have to be corrected, if you look from top to bottom, the entries are from small areas (airports) to bigger areas (regions) to continents or the whole world. The example above is ordered following that rule.

There are tools available like X-Organizer helping you to sort the entries, after sorting you should always check if the order is correct.

If you use the global VFR-Landmarks, VFR-Aerials or any other global scenery outside Europe, please set them **below X-EUROPE** to avoid doubled objects in Europe.

X-EUROPE

X-EUROPE has 7 different layers:

Layer 1-vfr

Includes objects from 3D warehouse plus corrections, WED xml file is included, one group for corrections and one group for each tile

Layer 2-regions

changes objects to typical houses in special regions, yet done for the Mediterranean, Alps region, Norway/Scandinavia, Greece and Northern Germany

Layer 3-extras

This is the layer having extra elements, deactivate this layer if your hardware is at the limit. This layer has objects like caravans on camping sites, allotments, additional trees in cities and along rivers, and forests in parks etc.. It isn't necessary, but it's nice to see 😊

Layer 4-scenery

This layer is the main scenery, includes all houses and buildings, landmark objects. In the library.txt is the definition which objects to use, see next page.

This layer is necessary, don't disable this layer!

Layer 5-forests

All forests, wetlands, vineyards and orchards from OSM, splitted up at motorways, streets and railways to avoid overlapping forests. This layer is generated in two ways,

1. for OSM areas with a script made by Oscar Pilote, forests are split at streets and railways and have an exclusion for forests from below over the whole tile.
2. For OSM+autogen areas, forests are generated by w2xp, having smart exclusions to prevent forests from below where a forest from OSM is generated. So, where OSM data is poor, forests from below are shown in the simulator.

Layer 6-autogen

For all tiles having low OSM data (OSM+autogen areas, see overview page 5), this layer has additional houses along residential streets. For these tiles layer 4 uses smart exclusions, showing this additional objects from this layer where OSM data is poor.

Layer 7-network

All network for roads, railways, power lines and aerals.

In the standard X-Plane sceneries, the network is processed to increase performance and reduce data. That means, that the number of reference points e.g. for a road is reduced which leads to more angular roads, which could have the effect shown in this screenshot:



Some roads are not exactly following the roads of the orthographic scenery, so you may have houses

intersecting streets. If you don't like it, activate this layer 7, you may have a performance drop, but the roads and railroads will then exactly follow the orthographic scenery.

Library.txt

X-EUROPE uses a library to easily change objects without new generation of the scenery. Using a text editor you could add, change or deactivate objects. This could be done for the whole scenery (Europe) (in layer 4) or for defined regions, e.g. the Alps or Norway (in layer 2).

The library system of X-Plane allows to define regions by a rectangular (south/east to north/west) or by a bitmap which achieves a more precise definition of a region as you could select single tiles in the bitmap. But it allows only to change the whole tile, not parts of it.

With the library.txt in layer **4-scenery** you also could change objects to your personal needs. E.g. if you don't like a special object, you could change it to another, or you could deactivate this object if you map it to a blank object.

You don't like sailing boats on lakes? Edit the library.txt, identify the used element "x-europe/landmarks/sailing_boats.for" and map it to "objects/blank.for". The element is still in the scenery, but it isn't shown.

Or you have better objects (if so, please mail me 😊), then identify the element and map it to your object which you have to copy into the scenery.

In layer **4-scenery** there are 2 different libraries provided:

- library.txt
This is the active file for X-Plane, default it is identical to library - standard.txt
- library - standard.txt
This is for standard or HDv4 sceneries, which will show special ground polygons. In default, it's the same as library.txt.
- library - orthos.txt:
Polygons for golf, beach, sand, grass, pool, soccer, tennis are set to "objects/blank.pol", blended out in the scenery.

The default library.txt is the same as "library - standard.txt". For changing to orthos version, copy the "library - orthos.txt" to "library.txt". The "library.txt" is the valid library for X-Plane.

X-Plane 10 compatibility

The layer 7-network uses the European network of X-Plane 11 (roads_EU.net). To be compatible with X-Plane 10 – and that's **only needed if you use layer 7-network**, you have to map "roads_EU.net" to "roads.net" of XP10. Simply do this by adding one single line to this file:

"X-Plane 10\Resources\default scenery\1000 roads\library.txt"

Edit this library.txt, search for roads.net (near line 210) and add this line:

EXPORT lib/g10/roads_EU.net roads.net

The result should look like this:

```
209 EXPORT lib/roads/SecOvrPrV1.obj objects/structures/SecOvrPrV1-1.obj
210 PUBLIC
211 EXPORT lib/g10/roads.net roads.net
212 EXPORT lib/g10/roads_EU.net roads.net
213 PRIVATE
```

Remember: If you update X-Plane 10, may be the change has to be done again.

Vegetation

New since v4.5: The vegetation library now is in an extra folder "simHeaven_Vegetation_Library" and available as separate download (<https://simheaven.com/simdownloads/vegetation-library/>)

This library was needed for the upcoming continental packages for Africa, America, Asia and Australia. "simHeaven_Vegetation_Library" is a global vegetation library using climate zones.

For more information, see the manual of "simHeaven_Vegetation_Library".

Fixing problems and issues

At the release of **X-EUROPE** there are no known issues. If you encounter issues, please first check if this is really caused by **X-EUROPE** or maybe by another scenery or add-on or plugin.

Additionally it would be helpful if you first look into openstreetmap.org or f4map.com if something is wrong or missing. **X-EUROPE** is generated 100% automatically, only layer 1-vfr is “hand-made”.

If any issues occur and they can be fixed easily, there will be an update available, please look at the FAQ on my website: <http://simheaven.com/faq/>

Tips and tricks

Some software settings could decrease performance of X-Plane. I’m doing support for several years, helping others to solve problems and to increase performance. The main things you should check:

- If you have an antivirus or internet security program, probably all files are scanned while loading, which increases loading time. Make an exception in the antivirus software, and disable scanning for read access for all used drives, paths and folders of your sceneries and for X-Plane itself.
- disable (or delete) all sceneries not needed. X-Plane loads all “active” sceneries even if exclusions blend out every scenery below, this needs loading time and memory.
- check scenery loading times in the log.txt to identify which package is causing the delay/issue, in this example the loading time for the hd mesh scenery is 17 times as long as the second:

```
2724 1:33:54.446 I/SCN: DSF load time: 16096 for file Custom Scenery/Global Airports/Earth nav data/+50+000/+50+006.dsf (0 tris, 0 skipped for C
2725 1:33:54.537 I/SCN: DSF load time: 88891 for file Custom Scenery/simHeaven_X-Europe-1-vfr+corr/Earth nav data/+50+000/+50+006.dsf (0 tris, C
2726 1:33:54.717 I/SCN: DSF load time: 215535 for file Custom Scenery/simHeaven_X-Europe-3-extras/Earth nav data/+50+000/+50+006.dsf (0 tris, 0
2727 1:33:54.900 I/SCN: DSF load time: 177338 for file Custom Scenery/simHeaven_X-Europe-4-forests/Earth nav data/+50+000/+50+006.dsf (0 tris, C
2728 1:33:55.430 I/SCN: DSF load time: 452195 for file Custom Scenery/simHeaven_X-Europe-5-scenery/Earth nav data/+50+000/+50+006.dsf (0 tris, C
2729 1:33:56.069 I/SCN: DSF load time: 584335 for file Custom Scenery/simHeaven_X-Europe-7-network/Earth nav data/+50+000/+50+006.dsf (0 tris, C
2730 1:34:06.423 I/SCN: DSF load time: 10259833 for file Custom Scenery/zxx_hd_europe_scenery4/Earth nav data/+50+000/+50+006.dsf (1452134 tris, C
```

- disable all AI aircrafts, every AI aircraft takes CPU power and performance. If you like it busy on airports or in the sky, better use an add-on like “X-Life” or “World Traffic”.
- check BIOS and GPU settings for energy savings and check if it is enabled. If so, this could decrease performance, so disable energy savings if you want to have maximum performance of your hardware.
- remember if you use orthographic sceneries, ONE tile has a total size of 8 GB for zoom level (ZL) 17, so imagine how much data is loaded into memory and VRAM. Same is if X-Plane loads tiles during flight. For that I recommend the use of a SSD, better the use of m.2 SSD (if possible) to ensure fast loading.

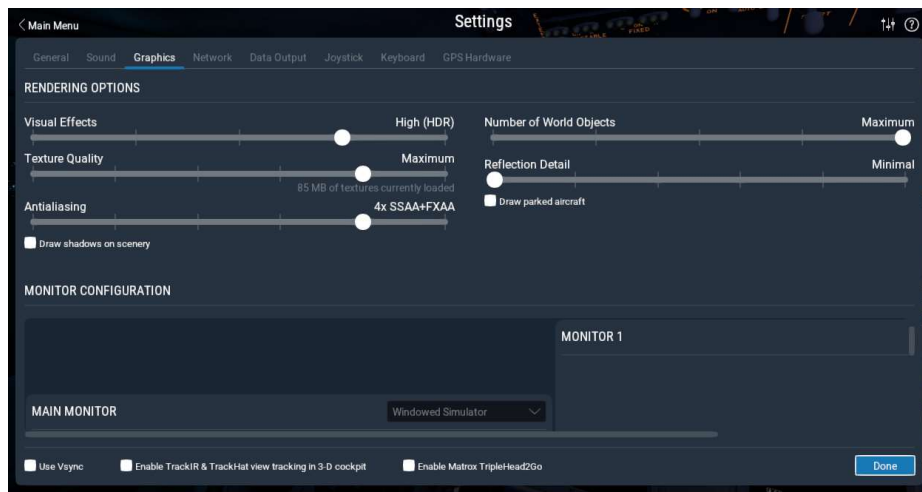
Hardware

There are many discussions about the “perfect” hardware, well, it depends on how much you want (or could) afford for flight simulation.

If you upgrade or buy a new computer, you have to check if the system in total and the single components fit together to avoid spending much money for a top graphic card and having the bottleneck elsewhere, like in the CPU or memory. I follow the talks in some threads about the perfect hardware, it’s more difficult to find the perfect “affordable” hardware.

X-Plane settings

Various settings influence the performance and look of X-Plane. If you have issues running **X-EUROPE**, first reduce number of world objects in the settings.



Graphic card settings

Check if the settings allow a high performance, if power management mode is set to maximum performance and even other settings.

Improve OSM

Nearly all of **X-EUROPE** is generated automatically using OSM data, the better the data the better the result. You could help to improve OSM data. You may discover things during flights which are missing or maybe wrong, you may see villages where OSM data lacks. If you like, add or change objects in OSM, see <https://www.openstreetmap.org>

Support

If you have any problems using **X-EUROPE**, send me an e-mail (in English/Deutsch/Español) to info@simHeaven.com , attaching log.txt, screenshots and if possible and needed the link to [OSM](https://www.openstreetmap.org) or [f4map.com](https://www.f4map.com).

This work was a great effort for me and my team, since beginning we spend thousands of hours to generate, test and improve the scenery. If you like the work, consider how much work it was for me and my team, consider the quality we achieved for the total area of **X-EUROPE** (1.700 tiles) bringing fun and realism to your flights for free.

Donation

If you want to support me and my work, help me covering costs for servers, software and plugins, please do a donation and say thanks to me and my team:

Donate to simHeaven: https://www.paypal.com/donate?hosted_button_id=RJRM2YX6R928Q
or use paypal.me/simheaven and „Send2Friend“ to avoid PayPal fees

Thank you, have fun flying,
Armin (aka PilotBalu)

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